

Name: I Forgot Again Team **Date:** 4/5/24 **Class:** Engineering

MyDesign Portfolio

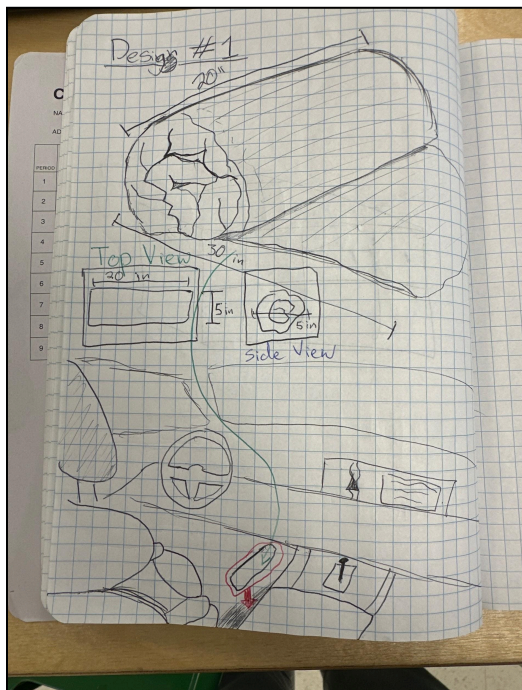
Element D: Design Concept Generation, Analysis, and Selection

Design Idea Generation (D1)

Design Idea #1

Date: 3/22/24

Description: A towel that can fill the gap and save items from dropping.

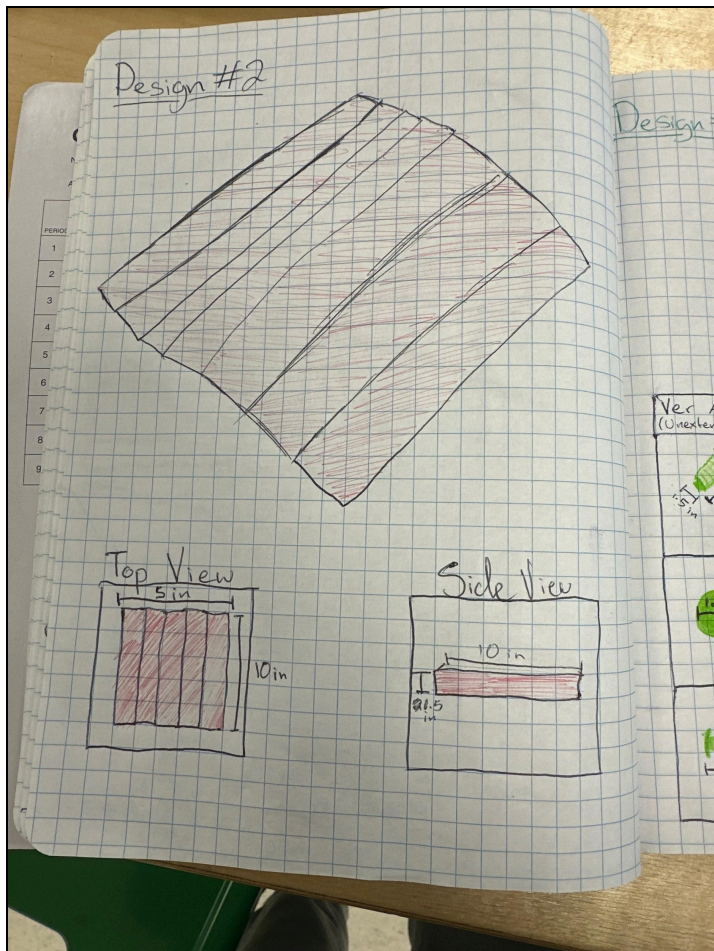


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Design Idea #2

Date: 4/2/24

Description: Hot Wheels tracks that can cover the gap from the top.

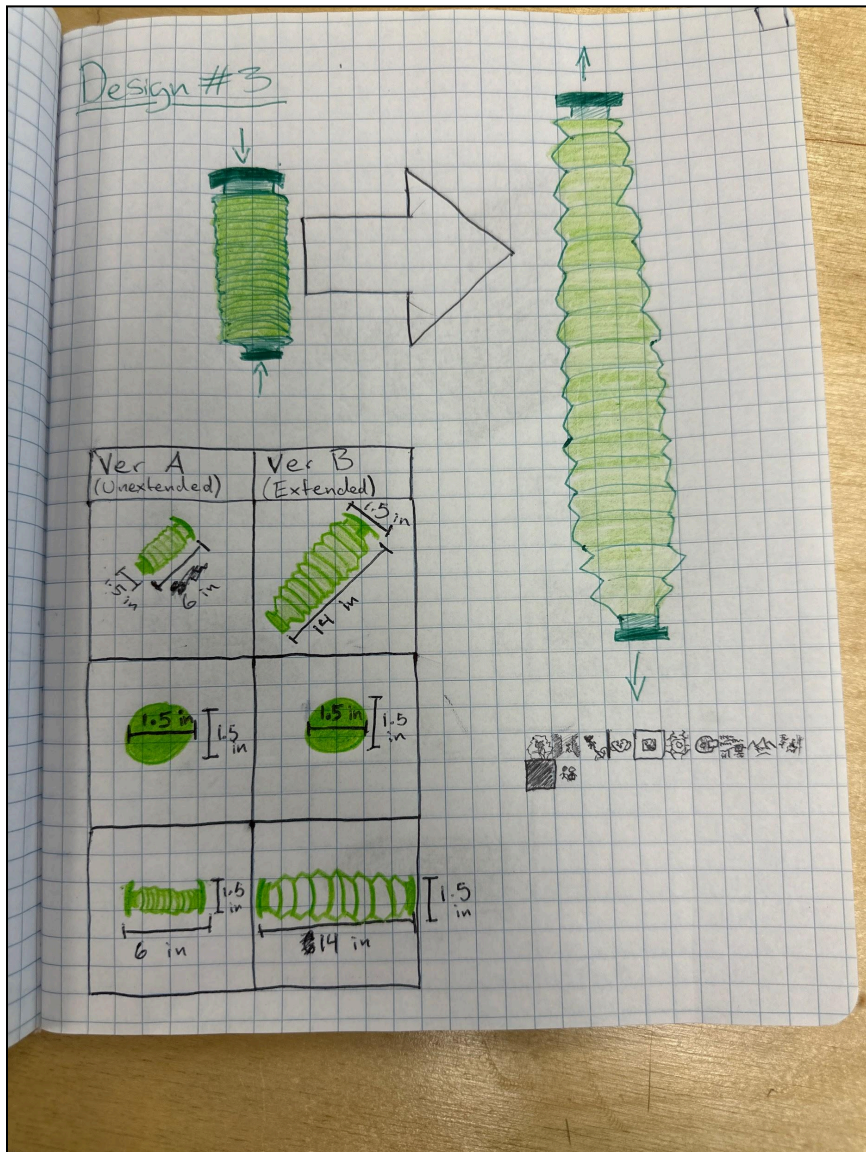


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Design Idea #3

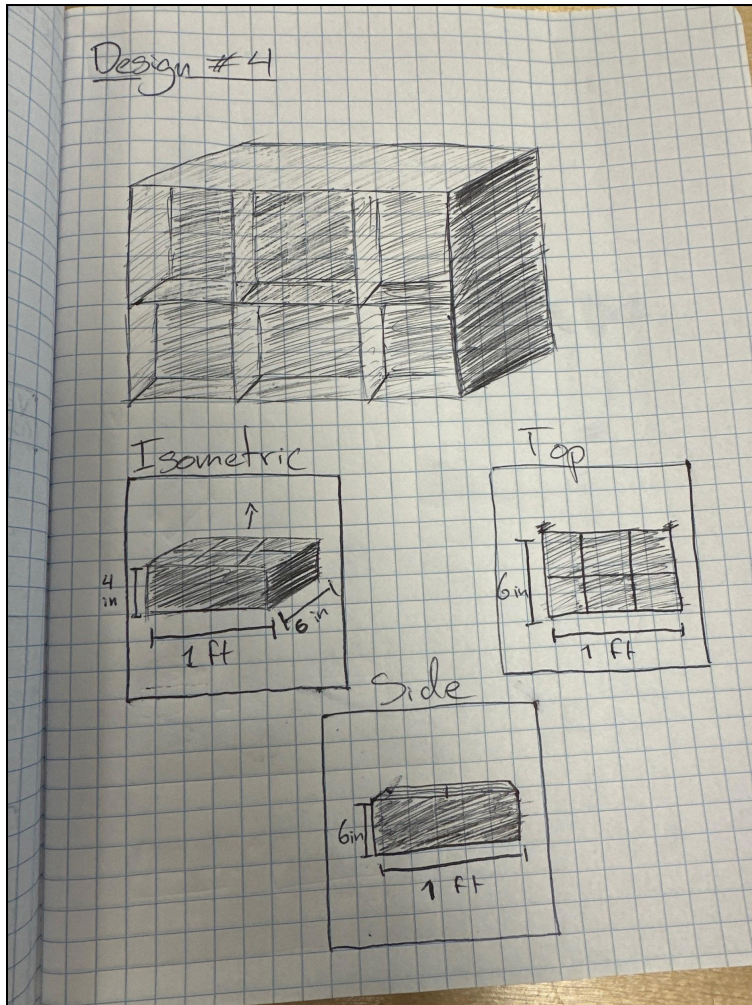
Date: 4/2/24

Description: We will stretch Pop Tubes to sit in the gap.



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Design Idea #4



Date: 4/2/24

Description: We will design compartments inside of the glove box of a car to hold items.

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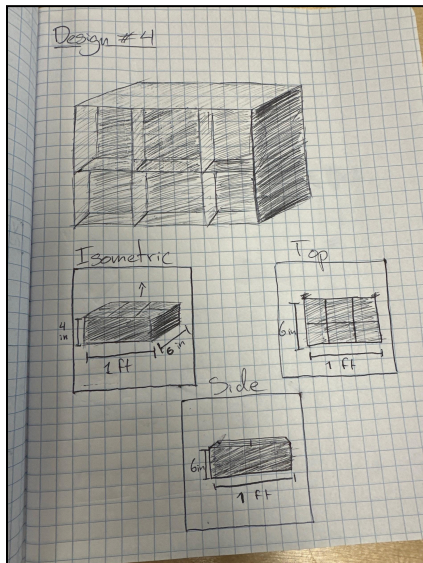
Design Idea Comparison and Selection (D1, D2)

Requirements *See Element C	Weight	Towel	Hot Wheels Track	Pop Tubes	Compartment nts
Universal Fit	20	10	10	10	20
Durability	20	15	10	10	20
Safety	20	10	10	10	20
score	100	35	30	30	60

Design Blueprint (D4)

Design Blueprint #1

Date: **04/11/24**



Manufacturing Methods:

Step 1: Measure out the sides of the glove box and or console in the car for it to fit properly in a glove box/compartment.

Step 2: Measure out the structure using the measurements of the compartment. We will make our dimensions

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Step 3: Cut the cardboard into vertical supports and horizontal shelves, then add half a centimeter wide slits going halfway through each piece at intersection points, then slide parts together, cut foam to fit snugly in the back of the shelves, then insert the shelves and adjust sizes if necessary, then proceed to insert foam, after which use the tape to lock the intersections and cover up weak spots in the cardboard.

Item	Cost each	# needed	Total
Cardboard	0\$	1 large sheet of cardboard	0\$
Foam (Styrofoam)	0\$	1 sheet of styrofoam	0\$
Hot glue	0\$	3 sticks of glue	0\$
Total Cost			0\$

Plan of Action (D3)

	A	B	C
1	Task	Team Member Assigned as Lead	Date Due.
2	Problem Scoping	B	3/18
3	Team Charter	B	3/21
4	Planing Guide	B	4/5
5	Element A	B	4/9
6	Element B	B	4/9
7	Element C	B	4/9
8	Element D	B	4/9
9	Element E	B	4/9
10	Element F	B	4/9

ITERATIONS:

4/15: Changed measurements from 12x6x4.5 inches to 10x5x4.5 inches to fix a car console. We created 4 compartments instead of 6 to fit the new size.

4/19: Started to iterate by adding a lid on top.



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Rubric

Element D: Design Concept Generation , Analysis and Selection						
	5	4	3	2	1	0
1. Process for Generating & Comparing Possible Solutions. The Process for Generating Possible Solutions _____.	was comprehensive, iterative, and consistently defensible (viable design highly likely)	was thorough, iterative, and generally defensible (viable design likely)	was adequate, generally iterative and defensible (viable design is possible)	was partial or overly general and only somewhat iterative and/or defensible (raising issues with the design viability)	was incomplete or minimally iterative and/or defensible (raising issues with the design viability)	was not evident that there was an attempt
2.Design Solution Choice. The Design Solution chosen _____.	was well-justified and demonstrated attention to all design requirements	was justified and demonstrated attention to most if not all design requirements	was explained with reference to at least some design requirements	was not sufficiently explained with reference to design requirements	lacked support related to design requirements	was not attempted or evidence not provided

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3. Plan of Action. The plan of action _____ 	<i>has considerable merit and would easily support repetition and testing for effectiveness by others</i>	<i>would support repetition and testing for effectiveness by others</i>	<i>might not clearly or fully support repetition and testing for effectiveness by others</i>	<i>has insufficient detail to allow for testing for replication of results</i>	<i>is lacking to the point that the design solution cannot be tested</i>	<i>cannot be executed or does not exist</i>
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